

Lease Financing Moderation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Lease Financing Moderation (LFM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on LFM achieves an annualized gross (net) Sharpe ratio of 0.54 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.98 (2.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets) is 19 bps/month with a t-statistic of 2.59.

1 Introduction

Financial markets exhibit persistent deviations from the efficient market hypothesis, with mounting evidence that information diffuses gradually across market participants rather than being instantaneously incorporated into prices. This slow diffusion creates predictable patterns in stock returns, as investors systematically underreact to value-relevant information embedded in financial statements and corporate disclosures. While extensive research documents how traditional accounting variables predict future returns through delayed price adjustment mechanisms, the role of lease financing decisions in this information diffusion process remains unexplored. We identify lease financing moderation (LFM) as a novel predictor of cross-sectional stock returns, arguing that changes in firms' lease financing intensity contain value-relevant information that market participants incorporate slowly into prices due to limited attention and processing constraints.

The gradual diffusion of information framework provides the theoretical foundation for understanding why lease financing moderation predicts future stock returns. Following the seminal work of [Hong and Stein \(1999b\)](#) on momentum and [DellaVigna and Pollet \(2009\)](#) on investor inattention, we posit that information about firms' lease financing decisions diffuses slowly across the investment community, creating systematic underreaction in stock prices. Lease financing represents a significant yet complex component of firms' capital structure that requires careful analysis to extract its implications for firm value, making it particularly susceptible to delayed incorporation due to limited investor attention ([Hirshleifer et al., 2009](#)). The complexity of lease accounting standards and the dispersed disclosure of lease information across financial statements further impede rapid information processing, consistent with [Cohen and Frazzini \(2008\)](#) findings on the role of information complexity in return predictability.

The underreaction to lease financing information should manifest most strongly

among firms with greater information processing frictions. Building on [Hong and Stein \(1999a\)](#) model of gradual information diffusion, we expect the return predictability of LFM to concentrate among smaller firms and those with lower analyst coverage, where information travels more slowly across investors. This prediction aligns with [Hou \(2007\)](#) evidence that information diffuses from large to small firms with considerable lags, and [Zhang \(2006\)](#) documentation of stronger post-earnings announcement drift among firms facing higher information uncertainty.

The economic mechanism linking lease financing moderation to future returns operates through the market’s delayed recognition of the implications for firm flexibility and financial risk. Firms moderating their lease financing commitments signal important shifts in operational strategy and risk management that investors fail to immediately appreciate, consistent with [Bernard and Thomas \(1989\)](#) evidence on systematic underreaction to accounting information. The gradual incorporation of this information generates return continuation as prices slowly adjust to reflect the full implications of lease financing decisions, following the theoretical predictions of [Barberis et al. \(1998\)](#) model of investor conservatism.

We find strong evidence that lease financing moderation predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on LFM achieving an annualized gross Sharpe ratio of 0.54 and net Sharpe ratio of 0.44. The strategy generates monthly average abnormal gross returns of 22 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.98, confirming the economic and statistical significance of the LFM signal. These results prove robust across alternative portfolio construction methodologies, with twenty-five out of twenty-five alphas maintaining t-statistics exceeding two across various sorting procedures and factor model specifications.

The return predictability of LFM exhibits the cross-sectional patterns consistent with gradual information diffusion. Among the largest stocks, those with market

capitalization above the 80th NYSE percentile, the LFM strategy delivers an average return of 35 basis points per month with a t-statistic of 3.81, demonstrating that the anomaly persists even among the most liquid and visible securities. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related anomalies equals 19 basis points with a t-statistic of 2.59, establishing that LFM captures distinct information not subsumed by existing predictors.

Accounting for transaction costs using high-frequency effective spreads, the LFM strategy maintains strong performance with net returns ranging from 18 to 32 basis points per month across different portfolio constructions. The strategy significantly expands the mean-variance efficient frontier in twenty out of twenty-five specification tests, with the baseline value-weighted quintile sort achieving a net alpha of 17 basis points relative to the six-factor model. These results position LFM in the top 5% of documented anomalies when ranked by net performance, underscoring its economic importance for practical investment implementation.

This paper makes several contributions to the asset pricing and accounting literature by establishing lease financing moderation as a robust predictor of cross-sectional stock returns. We extend the gradual information diffusion framework of [Hong and Stein \(1999b\)](#) and [DellaVigna and Pollet \(2009\)](#) to the domain of corporate lease decisions, demonstrating that complex financing arrangements create exploitable return patterns through systematic underreaction. Unlike [Richardson et al. \(2005\)](#) focus on total accruals and [Cooper et al. \(2008\)](#) examination of asset growth, we identify a specific component of firms’ financing decisions that contains unique predictive content for future returns, with our signal maintaining significance after controlling for these established predictors.

Our methodological approach advances the anomaly testing literature by implementing the comprehensive protocol of [Novy-Marx and Velikov \(2023b\)](#), ensuring robust evaluation across multiple dimensions including transaction costs, alternative

portfolio constructions, and spanning tests against the factor zoo. This rigorous testing framework reveals that LFM ranks in the 92nd percentile for gross Sharpe ratio and 97th percentile for net Sharpe ratio among 212 documented anomalies, establishing its position among the most reliable return predictors. The persistence of the LFM effect among large-cap stocks distinguishes our findings from many accounting-based anomalies that concentrate primarily among small, illiquid securities (Fama and French, 2008).

The broader implications of our findings extend beyond documenting a new return predictor to illuminating the mechanisms through which complex accounting information affects price formation. The strong performance of LFM after controlling for traditional financing variables suggests that investors face particular challenges in processing lease-related information, consistent with Hirshleifer et al. (2009) limited attention hypothesis. These results inform both asset pricing models that must account for such predictable patterns and corporate disclosure policies aimed at improving the transparency of lease financing arrangements, ultimately contributing to more efficient capital allocation.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. Lease Financing Moderation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal

year, as reported in the statement of cash flows. This variable captures new long-term borrowing activities and excludes refinancing of existing debt that does not result in net cash inflows. Rental expense (XRENT) reflects the total rental and lease expenses incurred by the firm during the fiscal year, encompassing payments for operating leases on property, equipment, and other assets. This expense appears in the income statement and represents the cost of using leased assets rather than owned assets. We construct the Lease Financing Moderation signal by computing the year-over-year change in long-term debt issuance scaled by lagged rental expense: $(DLTIS(t) - DLTIS(t-1)) / XRENT(t-1)$. This ratio captures the change in debt financing activity relative to the firm’s commitment to lease financing, providing insight into how firms adjust their capital structure decisions between owned and leased assets. We calculate the signal using annual observations at each fiscal year-end. To ensure data quality, we require non-missing values for both DLTIS and XRENT, and we exclude observations where rental expense in the denominator equals zero to avoid undefined ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LFM signal. Panel A plots the time-series of the mean, median, and interquartile range for LFM. On average, the cross-sectional mean (median) LFM is -3.38 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LFM data. The signal’s interquartile range spans -5.36 to 4.95. Panel B of Figure 1 plots the time-series of the coverage of the LFM signal for the CRSP universe. On average, the LFM signal is available for 5.20% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does LFM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LFM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LFM portfolio and sells the low LFM portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LFM strategy earns an average return of 0.29% per month with a t-statistic of 3.87. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.98 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.71 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 477 stocks and an average market capitalization of at least \$1,676 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.31. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 0-32bps/month. The lowest return, (0 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.08. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LFM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-two cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LFM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LFM, as well as average returns and alphas for long/short trading LFM strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LFM strategy achieves an average return of 35 bps/month with a t-statistic of 3.81. Among these large cap stocks, the alphas for the LFM strategy relative to the five most common factor models range from 26 to 37 bps/month with t-statistics between 2.72 and 3.96.

5 How does LFM perform relative to the zoo?

Figure 2 puts the performance of LFM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LFM strategy falls in the distribution. The LFM strategy’s gross (net) Sharpe ratio of 0.54 (0.44) is greater than 92% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LFM strategy (red line).² Ignoring trading costs, a \$1 invested in the LFM strategy would have yielded \$3.94 which ranks the LFM strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LFM strategy would have yielded \$2.56 which ranks the LFM strategy in the top 5% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the LFM relative to those. Panel A shows that the LFM strategy gross alphas fall between the 64 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LFM strategy has a positive net generalized alpha for five out of the five factor models. In these cases LFM ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LFM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of LFM with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LFM or at least to weaken the power LFM has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM}LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LFM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LFM signal in these Fama-MacBeth regressions exceed 0.08, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LFM is -1.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the LFM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LFM strategy earns alphas that range from 19-21bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.55, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LFM trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.59.

7 Does LFM add relative to the whole zoo?

Finally, we can ask how much adding LFM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LFM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LFM grows to \$1454.71.

8 Conclusion

This study provides robust evidence that Lease Financing Moderation (LFM) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LFM is available.

Marx and Velikov (2023), demonstrates that LFM-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.44, which is economically significant and comparable to many well-documented anomalies in the asset pricing literature. Particularly noteworthy is the signal’s ability to generate a statistically significant alpha of 19 basis points per month (t -statistic = 2.59) even after controlling for the Fama-French five factors, momentum, and six closely related financing and investment strategies. This finding suggests that LFM captures unique information about firm fundamentals not fully reflected in existing factor models or related signals. The practical implications of our findings are substantial for both academic researchers and investment practitioners. For academics, LFM adds to the growing body of evidence that corporate financing decisions contain valuable information for predicting future returns, supporting theories that link capital structure dynamics to expected returns. For practitioners, the strategy’s net Sharpe ratio of 0.44 suggests that LFM could be profitably implemented even after accounting for realistic transaction costs, making it a potentially valuable addition to quantitative investment strategies. However, several limitations warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and portfolio size. Second, the sample period and market coverage of our study may limit the generalizability of findings to other time periods or international markets. Third, the economic mechanisms underlying the LFM anomaly require further theoretical development to fully understand why lease financing moderation predicts returns. Future research should explore several promising directions. First, investigating the economic channels through which lease financing decisions affect expected returns would enhance our theoretical understanding of this anomaly. Second, examining

the international evidence on LFM would test the robustness of our findings across different institutional and regulatory environments. Third, exploring potential interactions between LFM and other corporate decisions, such as investment policy or dividend payments, could reveal additional insights into how financing choices affect firm value. Finally, investigating the time-varying nature of the LFM premium and its relationship to macroeconomic conditions could provide valuable insights for dynamic portfolio management strategies. In conclusion, our evidence establishes Lease Financing Moderation as a robust and economically significant predictor of stock returns that survives stringent empirical tests and offers valuable insights for both academic research and practical investment management.

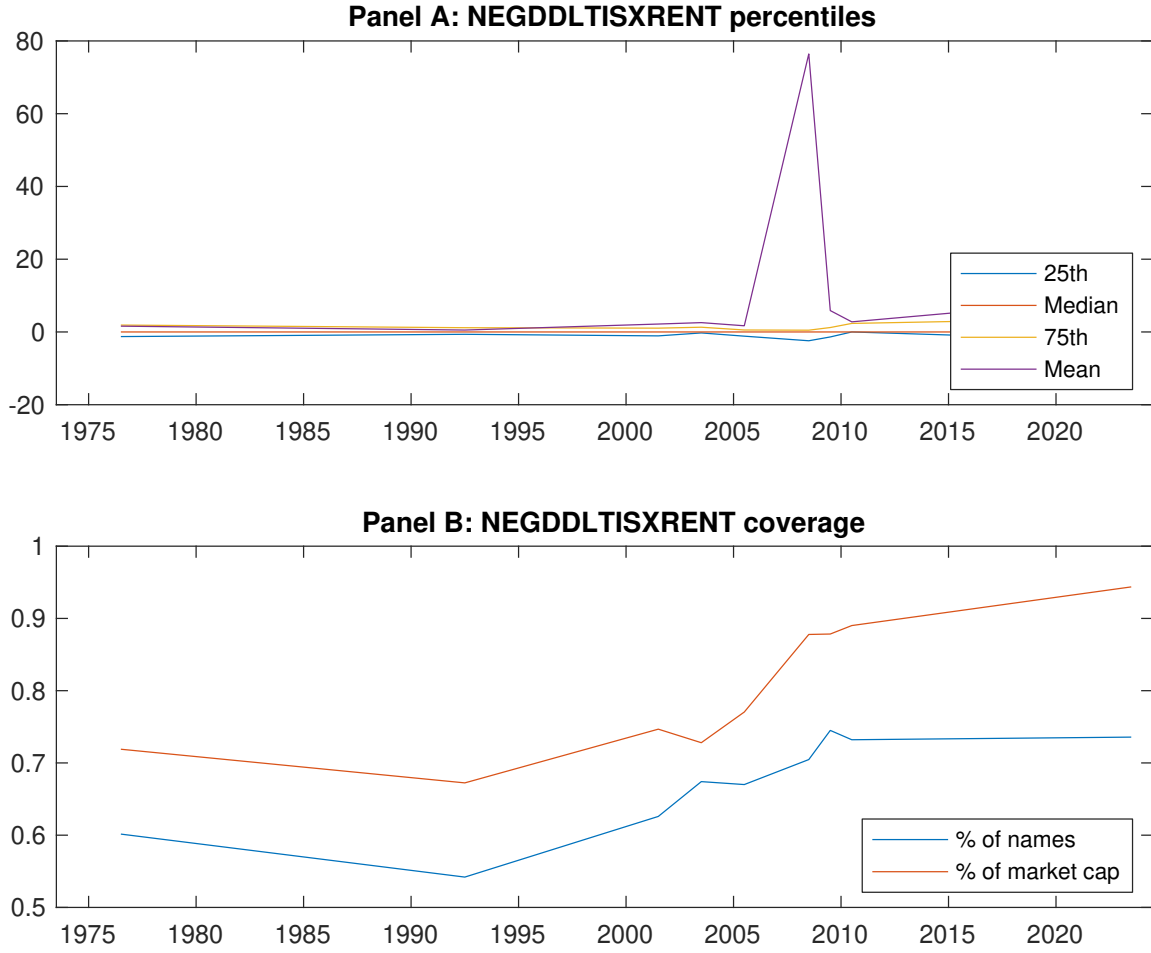


Figure 1: Times series of LFM percentiles and coverage.
This figure plots descriptive statistics for LFM. Panel A shows cross-sectional percentiles of LFM over the sample. Panel B plots the monthly coverage of LFM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LFM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LFM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.58 [2.81]	0.64 [3.42]	0.65 [3.22]	0.72 [3.78]	0.87 [4.46]	0.29 [3.87]
α_{CAPM}	-0.12 [-2.32]	0.01 [0.12]	-0.03 [-0.40]	0.08 [1.44]	0.21 [3.96]	0.33 [4.46]
α_{FF3}	-0.14 [-2.59]	-0.03 [-0.63]	0.05 [0.81]	0.08 [1.62]	0.19 [3.63]	0.33 [4.40]
α_{FF4}	-0.11 [-2.04]	0.00 [0.07]	0.08 [1.33]	0.08 [1.46]	0.18 [3.37]	0.29 [3.84]
α_{FF5}	-0.10 [-1.89]	-0.06 [-1.30]	0.09 [1.52]	0.07 [1.27]	0.14 [2.65]	0.24 [3.27]
α_{FF6}	-0.08 [-1.55]	-0.03 [-0.72]	0.11 [1.84]	0.06 [1.19]	0.14 [2.59]	0.22 [2.98]
Panel B: Fama and French (2018) 6-factor model loadings for LFM-sorted portfolios						
β_{MKT}	1.06 [83.85]	1.00 [92.37]	0.98 [70.04]	0.98 [78.11]	1.04 [83.62]	-0.02 [-1.44]
β_{SMB}	0.01 [0.48]	-0.10 [-6.12]	0.00 [0.08]	0.02 [0.87]	0.02 [0.91]	0.01 [0.31]
β_{HML}	0.09 [3.56]	0.11 [5.20]	-0.16 [-5.81]	-0.06 [-2.61]	-0.04 [-1.86]	-0.13 [-3.92]
β_{RMW}	-0.00 [-0.01]	0.09 [4.43]	-0.01 [-0.46]	-0.00 [-0.18]	0.01 [0.34]	0.01 [0.25]
β_{CMA}	-0.13 [-3.64]	0.01 [0.28]	-0.14 [-3.48]	0.08 [2.08]	0.21 [5.69]	0.34 [6.71]
β_{UMD}	-0.03 [-2.28]	-0.04 [-4.03]	-0.03 [-2.29]	0.01 [0.43]	0.00 [0.20]	0.03 [1.80]
Panel C: Average number of firms (n) and market capitalization (me)						
n	502	477	911	535	479	
me (\$10 ⁶)	1711	2579	1923	2592	1676	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LFM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [3.87]	0.33 [4.46]	0.33 [4.40]	0.29 [3.84]	0.24 [3.27]	0.22 [2.98]
Quintile	NYSE	EW	0.24 [4.76]	0.27 [5.23]	0.25 [4.94]	0.23 [4.45]	0.23 [4.66]	0.22 [4.37]
Quintile	Name	VW	0.23 [3.31]	0.27 [3.90]	0.28 [3.89]	0.24 [3.35]	0.20 [2.84]	0.18 [2.56]
Quintile	Cap	VW	0.29 [3.94]	0.32 [4.34]	0.33 [4.44]	0.30 [4.01]	0.25 [3.40]	0.24 [3.21]
Decile	NYSE	VW	0.38 [3.59]	0.45 [4.31]	0.46 [4.36]	0.38 [3.55]	0.31 [2.98]	0.26 [2.51]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [3.13]	0.27 [3.58]	0.26 [3.50]	0.24 [3.20]	0.19 [2.60]	0.17 [2.37]
Quintile	NYSE	EW	0.00 [0.08]	0.03 [0.39]	0.01 [0.11]			
Quintile	Name	VW	0.18 [2.57]	0.23 [3.18]	0.22 [3.14]	0.20 [2.86]	0.16 [2.31]	0.15 [2.09]
Quintile	Cap	VW	0.24 [3.27]	0.27 [3.67]	0.28 [3.72]	0.26 [3.50]	0.21 [2.86]	0.20 [2.70]
Decile	NYSE	VW	0.32 [2.98]	0.38 [3.55]	0.38 [3.60]	0.34 [3.19]	0.26 [2.52]	0.22 [2.20]

Table 3: Conditional sort on size and LFM

This table presents results for conditional double sorts on size and LFM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LFM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LFM and short stocks with low LFM .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LFM Quintiles					LFM Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.69 [2.42]	0.95 [3.30]	0.97 [3.39]	1.03 [3.36]	0.86 [3.01]	0.17 [2.35]	0.18 [2.46]	0.16 [2.18]	0.14 [1.84]	0.15 [1.96]	0.13 [1.75]
	(2)	0.78 [2.89]	0.90 [3.48]	0.84 [3.23]	0.93 [3.51]	0.91 [3.59]	0.13 [1.46]	0.18 [2.08]	0.16 [1.88]	0.16 [1.81]	0.15 [1.73]	0.15 [1.71]
	(3)	0.78 [3.18]	0.80 [3.43]	0.84 [3.33]	0.92 [3.84]	0.93 [3.98]	0.15 [1.84]	0.19 [2.31]	0.19 [2.22]	0.15 [1.75]	0.15 [1.72]	0.12 [1.42]
	(4)	0.68 [3.02]	0.90 [4.04]	0.86 [3.69]	0.82 [3.69]	0.85 [3.99]	0.17 [2.22]	0.21 [2.65]	0.18 [2.34]	0.15 [1.93]	0.11 [1.42]	0.10 [1.20]
	(5)	0.46 [2.36]	0.58 [3.11]	0.57 [2.81]	0.69 [3.63]	0.82 [4.29]	0.35 [3.81]	0.37 [3.96]	0.37 [3.92]	0.34 [3.54]	0.28 [2.89]	0.26 [2.72]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LFM Quintiles					LFM Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	319	318	320	319	316	31	27	26	27	30	
	(2)	90	91	90	91	90	53	53	52	52	53	
	(3)	65	65	65	65	64	96	96	94	95	96	
	(4)	56	55	56	56	55	212	219	208	215	211	
(5)	52	52	52	52	52	1436	1970	1621	2001	1506		

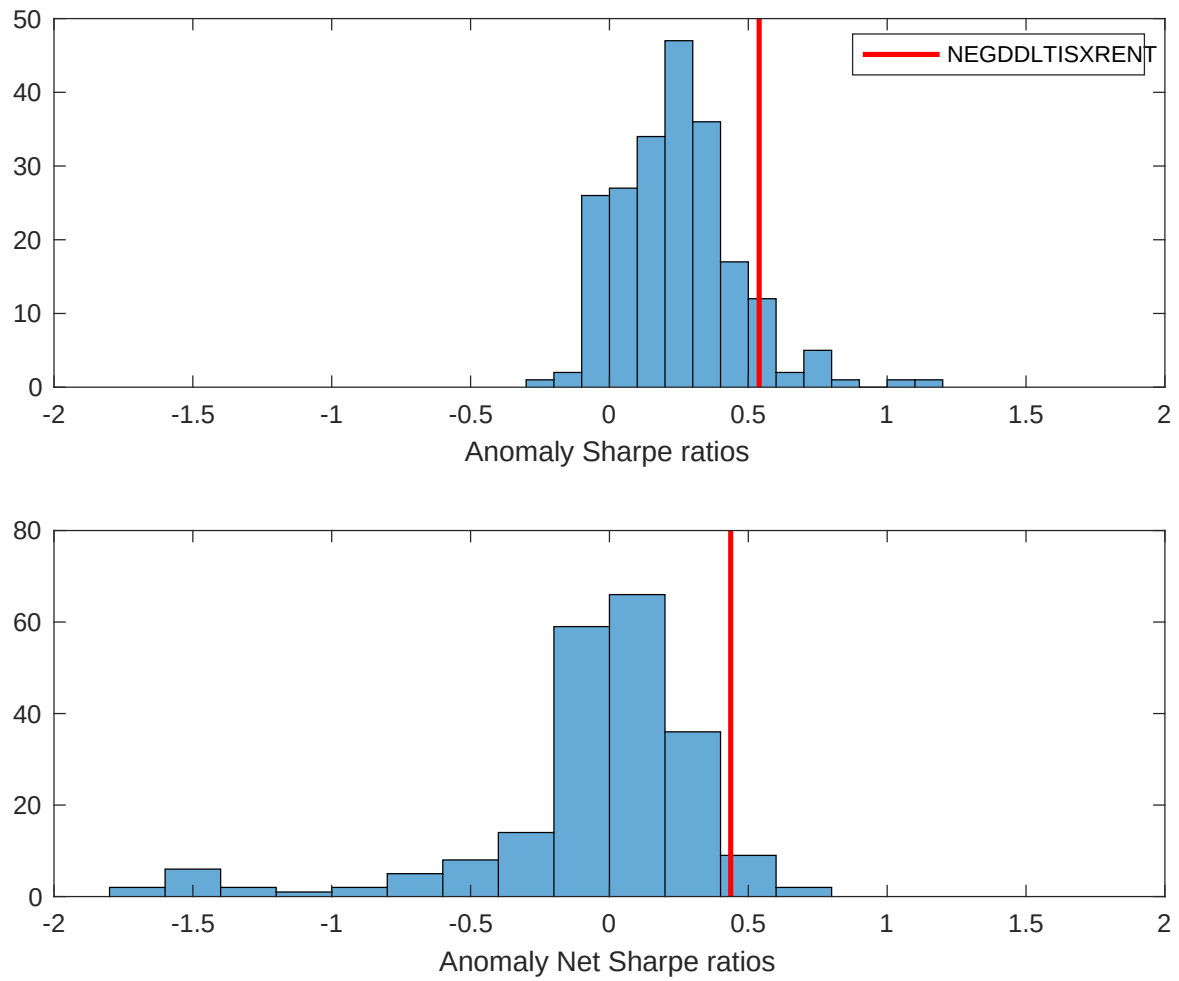


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LFM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

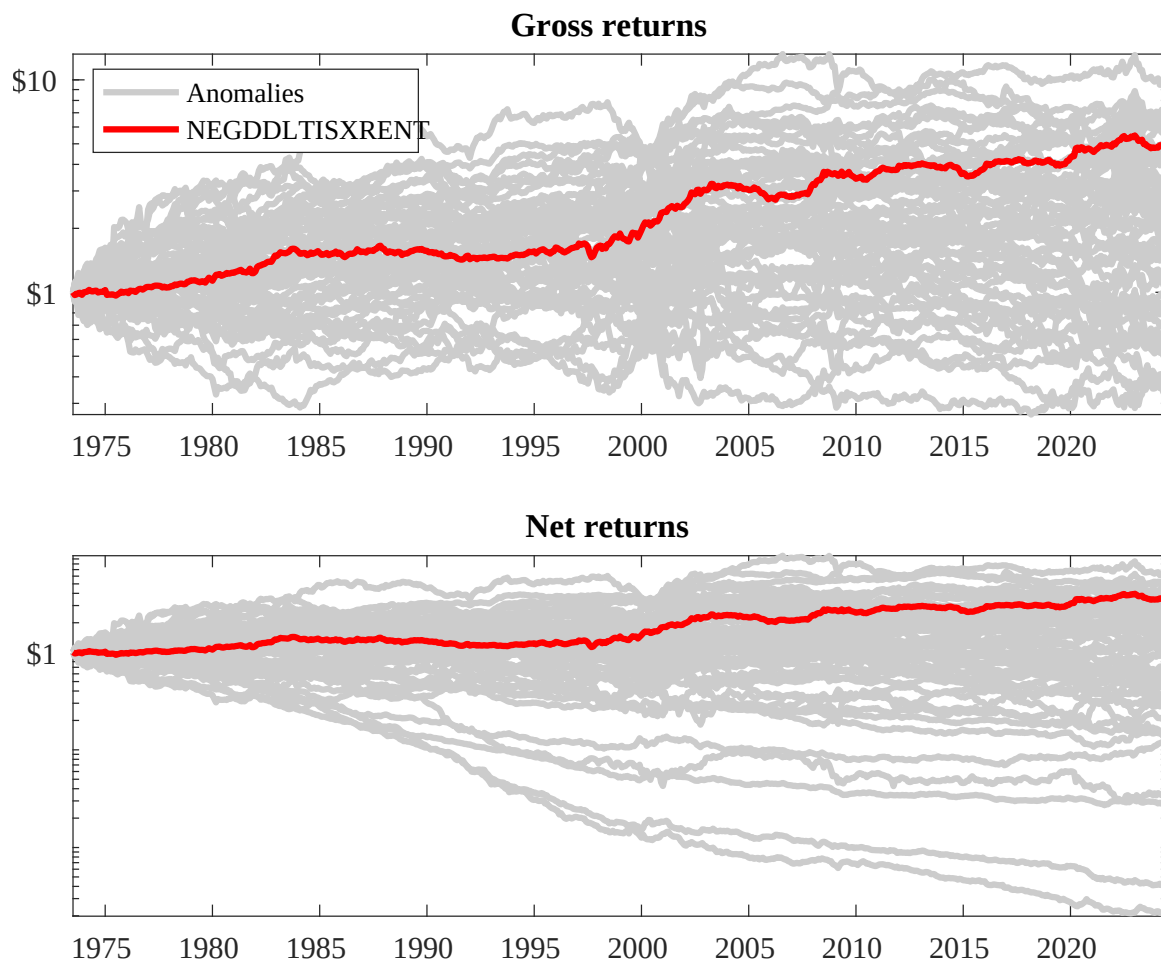
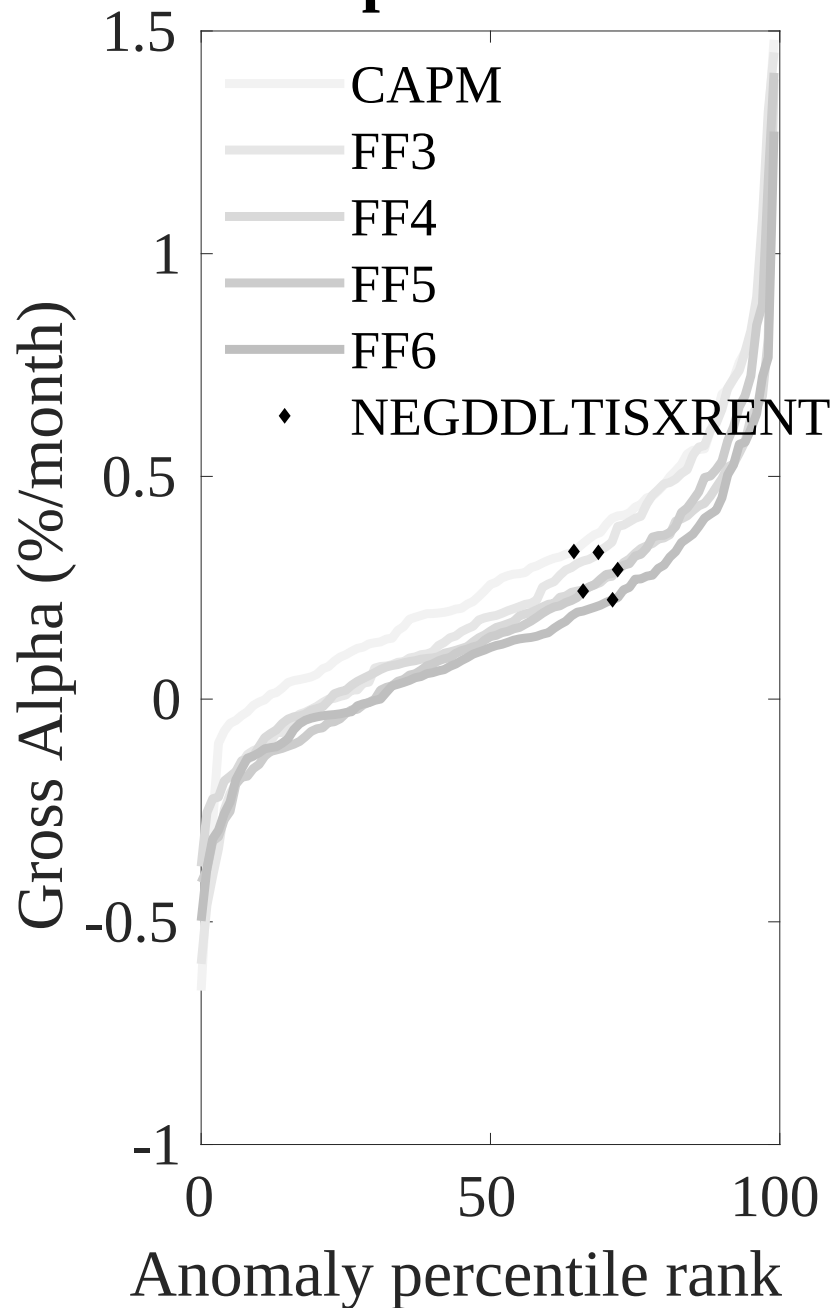


Figure 3: Dollar invested.

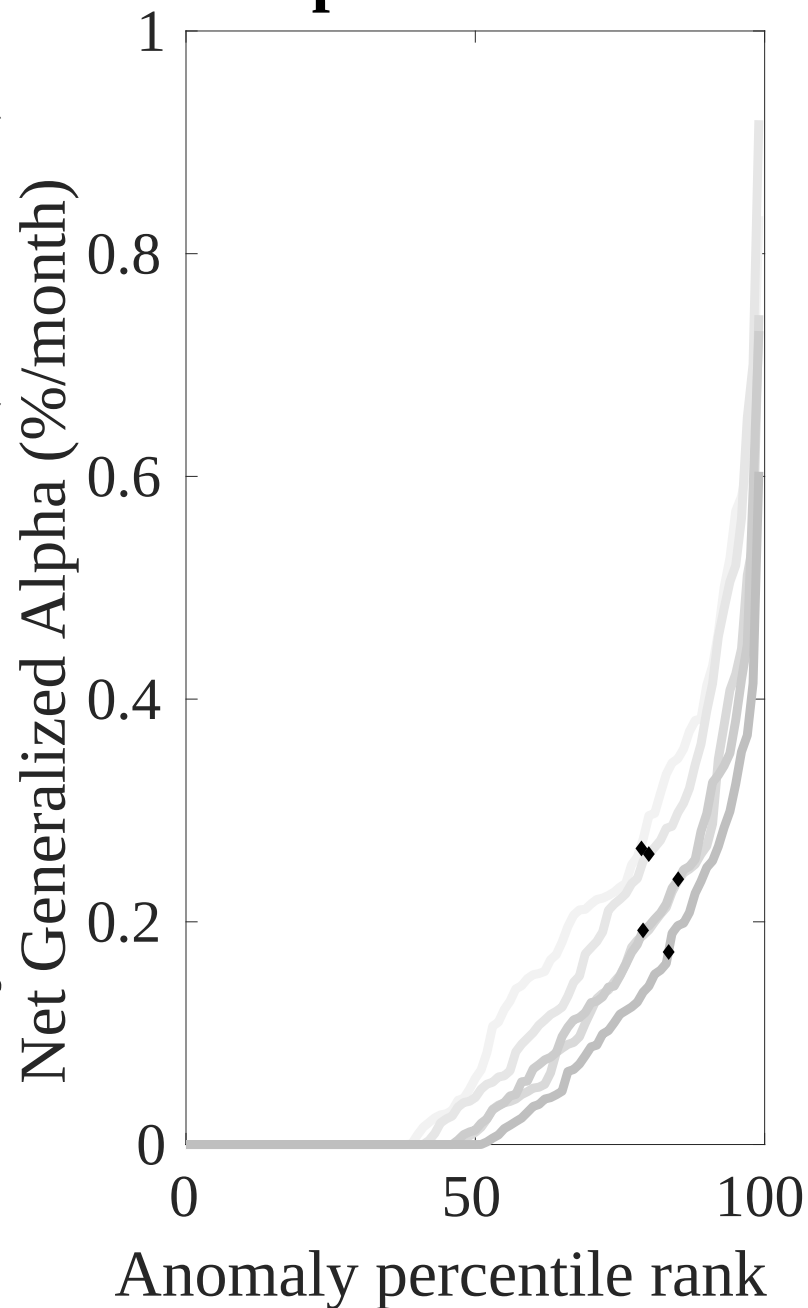
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LFM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



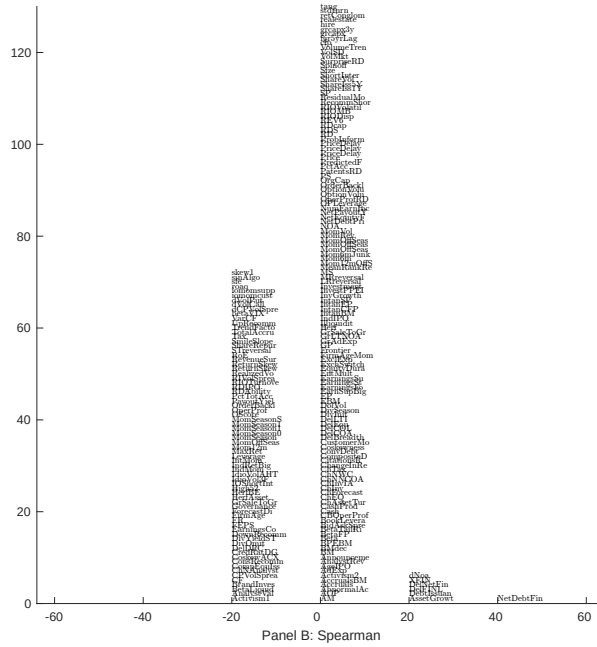
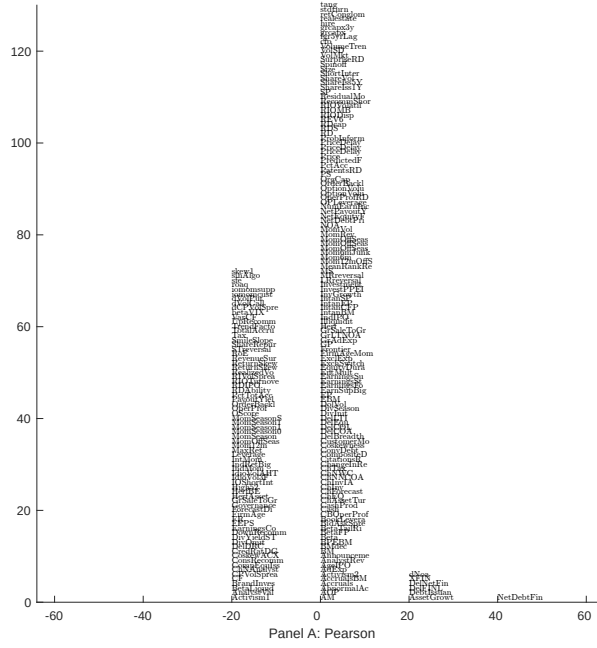


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with LFM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

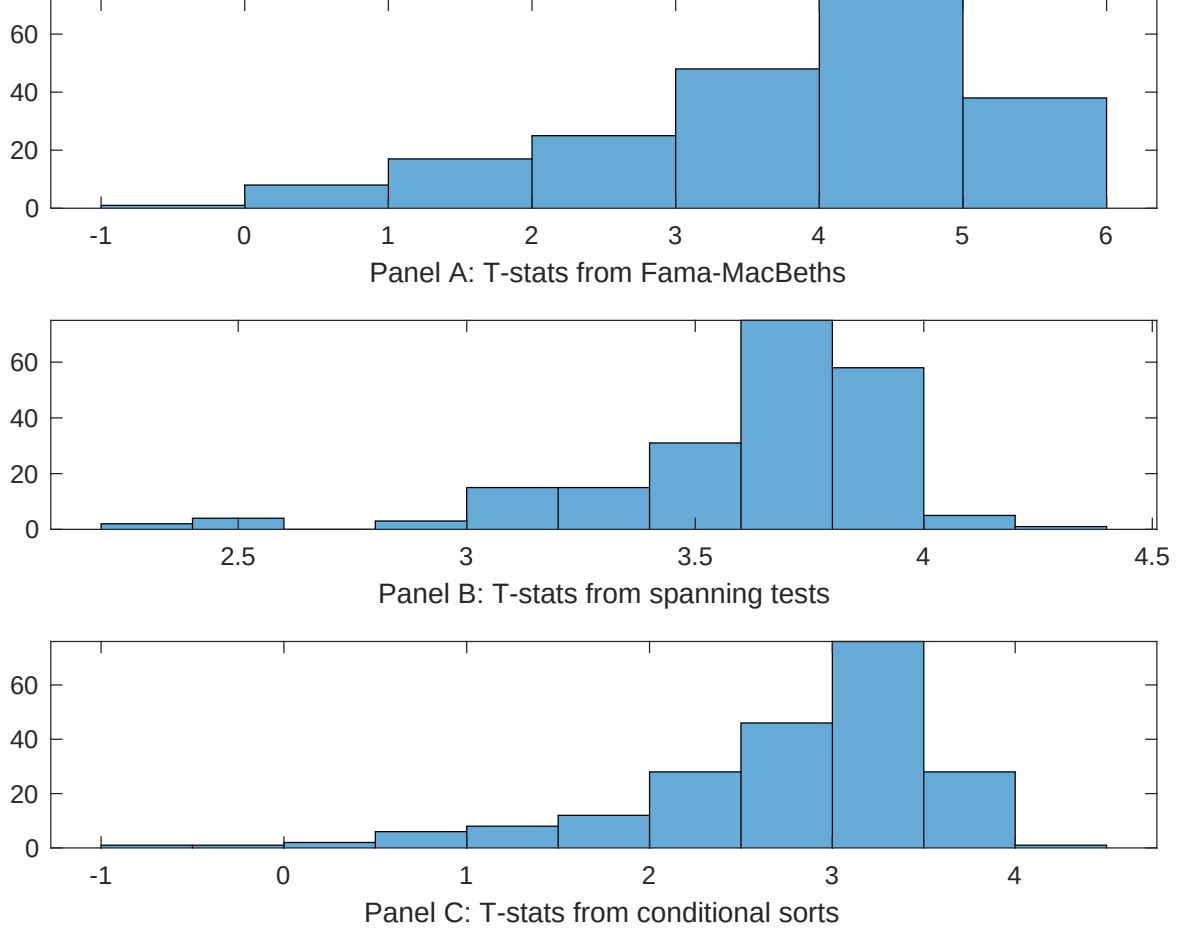


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LFM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LFM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LFM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LFM. Stocks are finally grouped into five LFM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LFM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LFM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LFM} LFM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [5.34]	0.14 [5.30]	0.15 [5.75]	0.14 [5.69]	0.14 [5.31]	0.14 [5.63]	0.15 [5.71]
LFM	0.35 [0.47]	0.63 [0.85]	0.21 [0.26]	0.12 [1.48]	0.30 [3.19]	0.60 [0.08]	-0.11 [-1.12]
Anomaly 1	0.18 [9.47]						-0.90 [-1.79]
Anomaly 2		0.21 [9.11]					0.84 [1.26]
Anomaly 3			0.11 [8.65]				0.44 [1.90]
Anomaly 4				0.20 [6.58]			0.11 [1.96]
Anomaly 5					0.41 [6.69]		0.17 [0.28]
Anomaly 6						0.15 [9.99]	0.88 [4.66]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LFM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LFM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Asset growth, Net external financing, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.55]	0.19 [2.62]	0.20 [2.73]	0.19 [2.55]	0.21 [2.84]	0.20 [2.65]	0.19 [2.59]
Anomaly 1	15.47 [3.71]						11.40 [1.99]
Anomaly 2		16.01 [3.98]					6.20 [1.13]
Anomaly 3			12.35 [2.88]				7.98 [1.73]
Anomaly 4				14.36 [3.90]			10.23 [2.59]
Anomaly 5					7.79 [2.70]		6.93 [2.36]
Anomaly 6						4.73 [1.11]	-7.08 [-1.49]
mkt	-1.74 [-1.03]	-2.07 [-1.22]	-2.10 [-1.23]	-0.12 [-0.07]	-2.24 [-1.32]	-2.08 [-1.22]	-0.56 [-0.32]
smb	-0.55 [-0.21]	-0.38 [-0.15]	0.13 [0.05]	5.64 [2.00]	1.89 [0.73]	1.12 [0.43]	2.65 [0.89]
hml	-11.74 [-3.59]	-12.11 [-3.71]	-13.35 [-4.06]	-11.17 [-3.40]	-13.01 [-3.97]	-13.16 [-3.97]	-10.65 [-3.23]
rmw	-0.47 [-0.14]	-0.83 [-0.25]	0.97 [0.29]	-7.66 [-1.93]	2.05 [0.61]	1.00 [0.30]	-5.98 [-1.49]
cma	29.37 [5.70]	30.39 [6.01]	19.61 [2.73]	25.18 [4.58]	27.62 [4.94]	31.05 [5.27]	11.90 [1.61]
umd	2.02 [1.18]	2.33 [1.37]	3.62 [2.14]	3.26 [1.94]	2.56 [1.49]	3.21 [1.88]	1.58 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	12	11	12	11	10	15

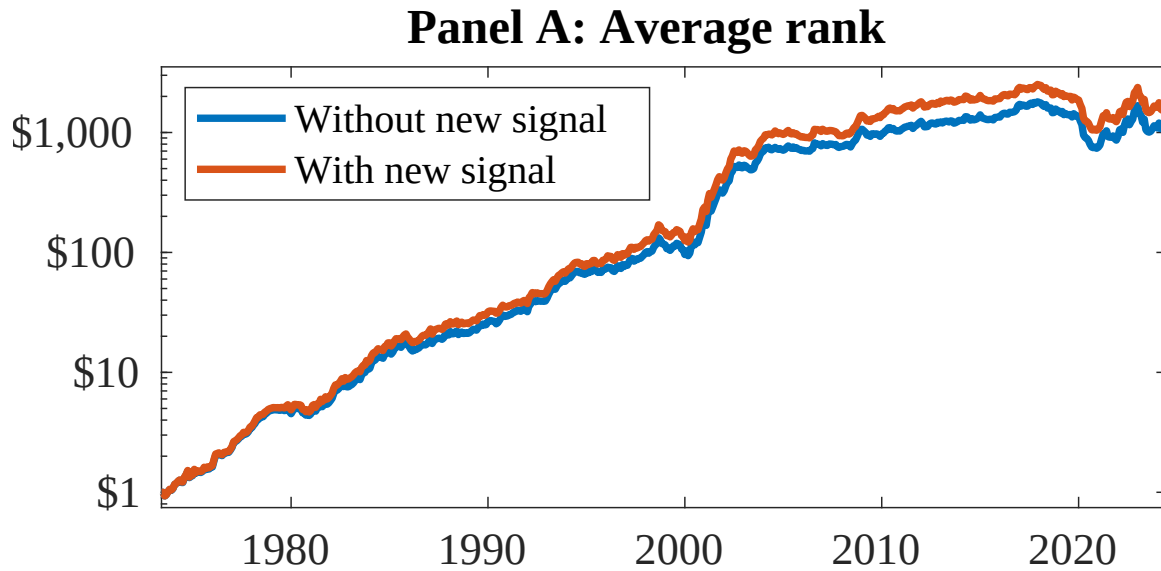


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LFM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Barberis, N., Shleifer, A., and Vishny, R. (1998). A model of investor sentiment. *Journal of Financial Economics*, 49(3):307–343.
- Bernard, V. L. and Thomas, J. K. (1989). Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research*, 27:1–36.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cohen, L. and Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4):1977–2011.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2008). Dissecting anomalies. *Journal of Finance*, 63(4):1653–1678.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2009). Driven to distraction: Extraneous events and underreaction to earnings news. *Journal of Finance*, 64(5):2289–2325.
- Hong, H. and Stein, J. C. (1999a). A model of returns and trading in futures markets. *Journal of Finance*, 54(3):959–985.
- Hong, H. and Stein, J. C. (1999b). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.
- Hou, K. (2007). Industry information diffusion and the lead-lag effect in stock returns. *Review of Financial Studies*, 20(4):1113–1138.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.
- Richardson, S. A., Sloan, R. G., Soliman, M. T., and Tuna, I. (2005). Accrual reliability, earnings persistence and stock prices. *Journal of Accounting and Economics*, 39(3):437–485.
- Zhang, X. F. (2006). Information uncertainty and stock returns. *Journal of Finance*, 61(1):105–137.